Spatiotemporal Imputation with Graph-Informed Flow Matching

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GiFlow uses a graph-informed prior to reduce transport cost by 65.19% and achieve faster, more accurate spatiotemporal imputation.

Graph-Informed Prior

- Diffusion models for spatiotemporal imputation start from an isotropic Gaussian; they face the problems of [1]:
  - absence of the available spatiotemporal structure
  - long transport path
  - slow sampling
- Let $\mathbf{X}_1^M = \mathbf{X}_1 \circ \mathbf{M}$ denote the observable spatiotemporal signal with $\mathbf{X}_1$ being the complete signal. The goal is to model the conditional distribution via

$$p_{\boldsymbol{\theta}}(\mathbf{X_1} \mid \mathbf{X}_1^M) = \int p_{\boldsymbol{\theta}}(\mathbf{X_1} \mid \mathbf{X}_0, \mathbf{X}_1^M) q_0(\mathbf{X}_0 \mid \mathbf{X}_1^M)\, d\mathbf{X}_0.$$

  Setting q_0 to a problem-agnostic Gaussian neglects the spatiotemporal structure.
- We aim to have a more efficient generation by constructing more structured priors.
- We use continuous spatiotemporal filtering to construct the graph-informed prior:

$$\mathbf{X}_{\boldsymbol{\tau}} = e^{-\tau_\eta \mathbf{L}_\eta}\, \mathbf{X}_1^M\, e^{-\tau_\xi \mathbf{L}_\xi}.$$

- From the perspective of minimizing transport cost, we obtain (τ_η, τ_ξ) by solving:

$$\underset{\tau_\eta, \tau_\xi > 0}{\text{minimize}} \ \left\| \mathbf{X}_1 - e^{-\tau_\eta \mathbf{L}_\eta} \mathbf{X}_1^M e^{-\tau_\xi \mathbf{L}_\xi} \right\|^2 + \alpha_\tau \mathrm{tr}\Big(\big(e^{-\tau_\eta \mathbf{L}_\eta} \mathbf{X}_1^M e^{-\tau_\xi \mathbf{L}_\xi}\big)^\top \mathbf{L}_\eta\, e^{-\tau_\eta \mathbf{L}_\eta} \mathbf{X}_1^M e^{-\tau_\xi \mathbf{L}_\xi} \Big)$$

Pipeline

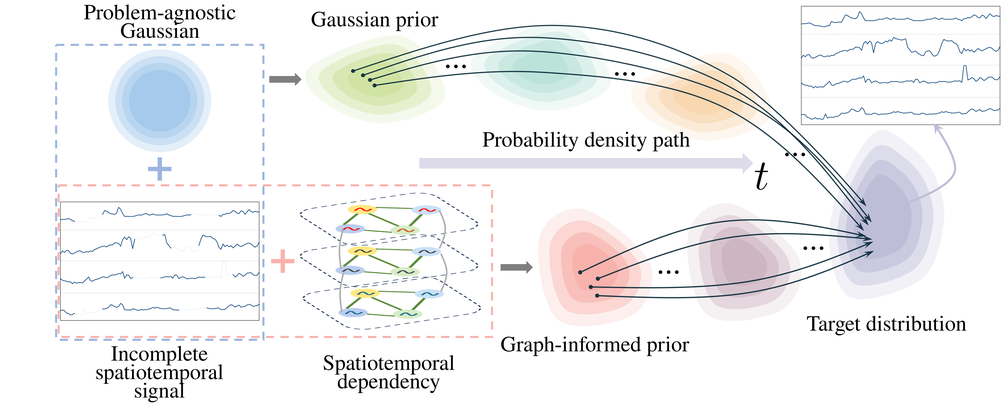


Figure GiFlow starts from a graph-informed prior instead of Gaussian noise, which yields shorter transport, better imputation, and faster inference

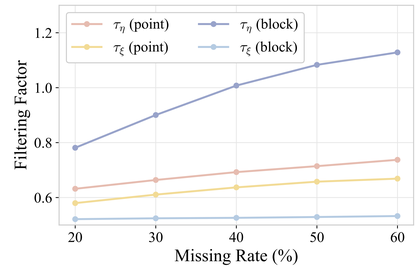
- To build a flow matching model for imputation tasks, it suffices to specify a conditional probability path and a vector field [2]. GiFlow uses a linear conditional probability path and a parameterized vector field modeled with a spatiotemporal model.

Theoretical Analysis & Empirical Evidence

Proposition 1. *Let C_s and C_t denote the spectral radii of $\mathbf{L}_\eta$ and $\mathbf{L}_\xi$, respectively. Then the truncation error with truncated spatial hops K_η and temporal hops K_ξ*

$$\left\| \mathbf{X}_{\boldsymbol{\tau}} - \left(\sum_{k=0}^{K_\eta} \frac{(-\tau_\eta)^k}{k!} \mathbf{L}_\eta^k \right) \mathbf{X}_1^M \left(\sum_{m=0}^{K_\xi} \frac{(-\tau_\xi)^m}{m!} \mathbf{L}_\xi^m \right) \right\|$$

is bounded by a constant depending on $\tau_\eta, \tau_\xi, K_\eta$, and K_ξ, which can be reduced either by decreasing (τ_η, τ_ξ) or increasing (K_η, K_ξ).



- As missing rate increases, we have larger τ_η, τ_ξ, indicating larger receptive fields.

Theorem 1. *Denote by p_0^{Gauss} and p_0^{G} the Gaussian prior and the graph-informed prior, where (τ_η, τ_ξ) are obtained via the minimization problem with $\alpha_\tau = 0$. The model with p_0^{G} has no larger transport cost along the probability path than p_0^{Gauss}:*

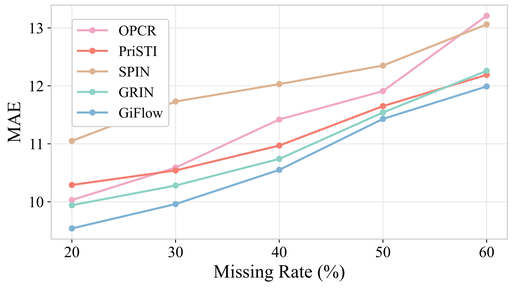
$$\mathcal{C}_{\mathrm{FM}}(p_0^{\mathrm{G}} \to q_1) \le \mathcal{C}_{\mathrm{FM}}(p_0^{\mathrm{Gauss}} \to q_1).$$

Model	T. C.	MAE	RMSE	MAPE
FM-Gauss	299.62	$12.79_{\pm 0.63}$	$22.15_{\pm 1.18}$	$26.85_{\pm 1.92}$
TFM	123.39	$10.12_{\pm 0.13}$	$19.60_{\pm 0.72}$	$22.41_{\pm 0.82}$
GFM	115.05	$9.75_{\pm 0.23}$	$18.67_{\pm 0.72}$	$21.55_{\pm 0.54}$
GiFlow	104.29	$\mathbf{9.54}_{\pm \mathbf{0.18}}$	$\mathbf{18.10}_{\pm \mathbf{0.78}}$	$\mathbf{21.27}_{\pm \mathbf{0.33}}$

- GiFlow with graph-informed prior reduces the transport cost by 65.19%, enabling more efficient generation and improving imputation accuracy.

Experiments and Results

Model	Air-36			AQI		
	MAE	RMSE	MAPE	MAE	RMSE	MAPE
Linear	$11.02_{\pm 0.12}$	$21.28_{\pm 0.60}$	$27.68_{\pm 0.50}$	$8.97_{\pm 0.04}$	$19.95_{\pm 0.25}$	$21.42_{\pm 0.11}$
KNN	$20.33_{\pm 0.14}$	$34.25_{\pm 0.71}$	$41.74_{\pm 0.96}$	$18.95_{\pm 0.03}$	$33.03_{\pm 0.27}$	$52.44_{\pm 0.20}$
FP	$16.51_{\pm 0.17}$	$28.68_{\pm 0.80}$	$38.51_{\pm 1.70}$	$15.65_{\pm 0.02}$	$27.20_{\pm 0.41}$	$45.70_{\pm 0.18}$
BRITS	$14.23_{\pm 0.17}$	$24.64_{\pm 0.59}$	$31.42_{\pm 0.72}$	$16.55_{\pm 0.12}$	$26.78_{\pm 0.36}$	$41.25_{\pm 0.35}$
SAITS	$14.32_{\pm 0.13}$	$23.85_{\pm 0.61}$	$31.62_{\pm 0.85}$	$17.95_{\pm 0.07}$	$28.95_{\pm 0.41}$	$44.89_{\pm 0.30}$
SPIN	$11.05_{\pm 0.87}$	$20.97_{\pm 1.90}$	$22.26_{\pm 1.56}$	$8.72_{\pm 0.12}$	$19.61_{\pm 0.67}$	$18.55_{\pm 0.28}$
GRIN	$9.94_{\pm 0.12}$	$19.09_{\pm 0.87}$	$21.95_{\pm 0.22}$	$7.97_{\pm 0.08}$	$18.46_{\pm 0.54}$	$16.81_{\pm 0.24}$
OPCR	$10.03_{\pm 0.12}$	$19.32_{\pm 0.60}$	$21.61_{\pm 0.68}$	$8.40_{\pm 0.20}$	$19.30_{\pm 0.69}$	$16.91_{\pm 0.59}$
PriSTI	$10.29_{\pm 0.14}$	$19.66_{\pm 0.25}$	$21.91_{\pm 0.66}$	$8.17_{\pm 0.28}$	$19.85_{\pm 0.28}$	$16.37_{\pm 0.59}$
CoSTI	$9.95_{\pm 0.10}$	$19.66_{\pm 0.27}$	$22.08_{\pm 0.37}$	$8.10_{\pm 0.13}$	$18.59_{\pm 0.37}$	$17.17_{\pm 0.41}$
GiFlow	$\mathbf{9.54}_{\pm \mathbf{0.18}}$	$\mathbf{18.10}_{\pm \mathbf{0.78}}$	$\mathbf{21.27}_{\pm \mathbf{0.33}}$	$\mathbf{7.83}_{\pm \mathbf{0.10}}$	$\mathbf{17.80}_{\pm \mathbf{0.28}}$	$\mathbf{16.24}_{\pm \mathbf{0.31}}$



- GiFlow achieves better spatiotemporal imputation compared with other baselines.

	Air-36	AQI	PeMS08	S-10	S-20	S-30	S-50
Time	0.19	0.36	0.46	5.75	11.56	23.98	69.85
Memory	0.02	0.04	0.03	1.01	3.49	7.47	19.88

- The optimization of (τ_η, τ_ξ) is efficient and scalable.

Model	Air-36	AQI	PeMS08
PriSTI	9.30	43.12	7.46
CoSTI	0.37	8.41	3.63
GiFlow	0.28	2.47	0.99

- GiFlow demonstrates faster inference than the diffusion model [1] and the consistency model [3].

Conclusion

- We propose the first graph-informed generative approach for spatiotemporal imputation, replacing the conventional Gaussian prior with a graph-informed prior.
- We show theoretically that the graph-informed prior reduces the transport cost to the target distribution, leading to shorter and simpler transport trajectories.

References:
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